Image Formation With Lenses and Spherical Mirrors

SYNOPSIS

A curved mirror needs no new equation. The thin lens equation and the magnification equation both carry over unchanged — all that changes is which way the signs point, and one short relation between the radius of a mirror and its focal length. This note is that relation and those sign tables, and it is the note to keep open during every problem that follows.

BEFORE YOU START — PREDICT

A concave mirror is ground to a radius of curvature of **40 cm**. Where is its focal point? Write a number down. Most people's first answer is 40.

PERFORMANCE EXPECTATIONS

1. **Relate** the radius of curvature of a spherical mirror to its focal length.
2. **Apply** the mirror and thin lens equation $1/d_o + 1/d_i = 1/f$ to a mirror.
3. **Read** the sign convention table correctly for concave and convex mirrors and for converging and diverging lenses.
4. **Predict** whether a given arrangement gives a real or virtual, upright or inverted image, *before* computing anything.

4. Image formation with lenses and spherical mirrors

$$R = 2f$$

The centre of curvature sits at twice the focal length. Grind a mirror to radius R and you have fixed its focal length at $R/2$ — there is no further choice in the matter.

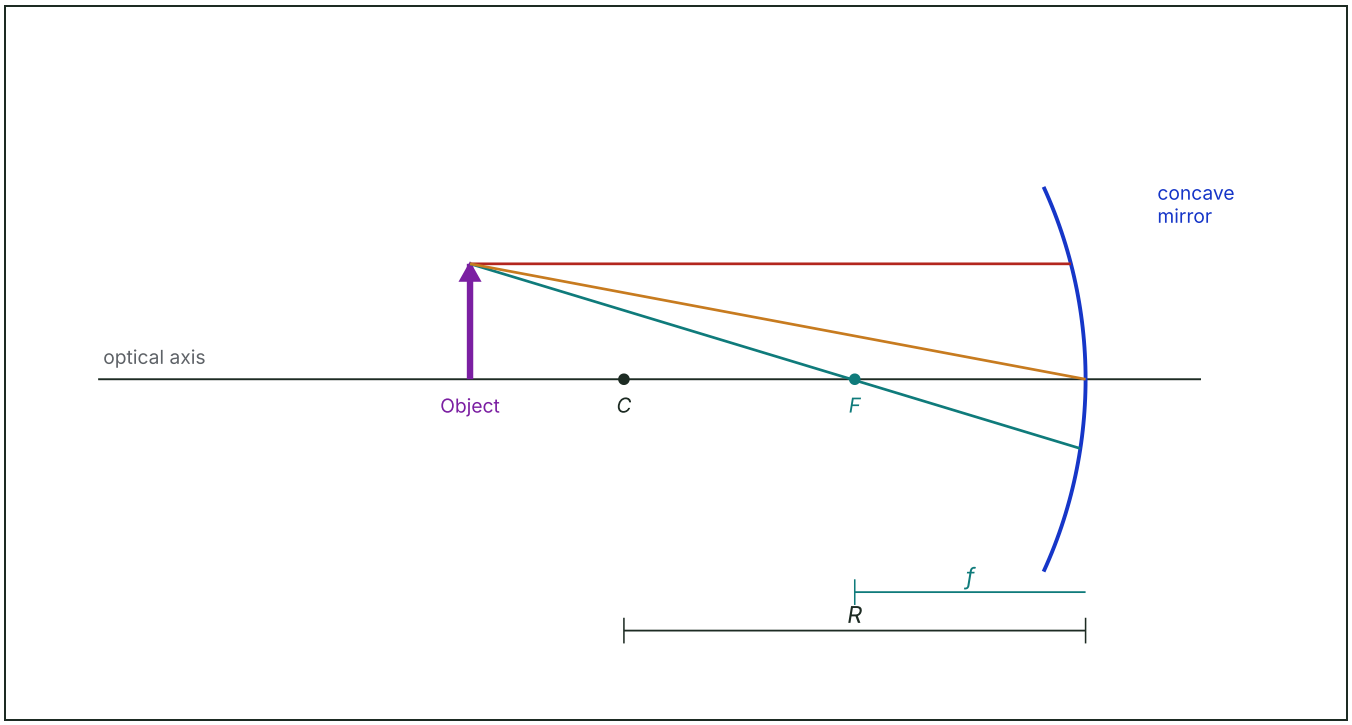


Figure 1. A concave mirror, object outside *C*. The arc is a real piece of a circle of radius *R* centred on *C*, and *F* sits exactly halfway along it. The three reflected rays cross between *F* and *C* because $1/d_o + 1/d_i = 1/f$ says they must.

Your job on Figure 1. The mirror, *C*, *F*, the object and the three incident rays are drawn for you. Reflect each ray by its own rule and mark where they cross. Before you measure anything: will the image land inside *F*, between *F* and *C*, or beyond *C*?

Sign convention for spherical mirrors and thin lenses

Applies to the mirror and thin lens equation $1/d_o + 1/d_i = 1/f$ and to the magnification equation $h_i/h_o = -d_i/d_o$.

	Spherical mirrors	Lenses
Focal length (f)	+ for concave mirrors	+ for a converging lens
	− for convex mirrors	− for a diverging lens
Object distance (d_o)	+ if the object is in front of the mirror (real object)	+ if the object is to the left of the lens (real object)
	− if the object is behind the mirror (virtual object)*	− if the object is to the right of the lens (virtual object)*
Image distance (d_i)	+ if the image is in front of the mirror (real image)	+ for a real image formed to the right of the lens by a real object
	− if the image is behind the mirror (virtual image)	− for a virtual image formed to the left of the lens by a real object

	Spherical mirrors	Lenses
Magnification (*m*)	+ for an image upright with respect to the object − for an image inverted with respect to the object	+ for an image upright with respect to the object − for an image inverted with respect to the object

* An optical system using several mirrors or lenses sometimes uses the image formed by the first as the object for the second. When that image falls on the far side, the object distance is negative and the object is called **virtual**. That is exactly the situation Note 9 is about.

The same thing as a quick table

	Mirror		Lens	
	Concave	**Convex**	**Concave**	**Convex**
Object (*o*)	+ Left	+ Left	+ Left	+ Left
Focal point (*f*)	+ Left	− Right	− Left	+ Right
Image (*i*)	+ / −	− Right	− Left	+ / −
Magnification (*m*)	+ / −	+	+	+ / −

The two columns with **+ / −** in them are the only ones where the answer depends on where you put the object; everywhere else the sign is fixed by the optic alone. **The convex lens is the one to know cold** — it is the case the rest of this module keeps coming back to, and the one with the most room to go wrong.

Worked example

1. A light bulb stands **0.75 m** in front of a converging lens of focal length **0.50 m**. A screen is moved back and forth behind the lens until the bulb comes into focus on it. **Where does the screen have to be, and what does the picture on it look like?**

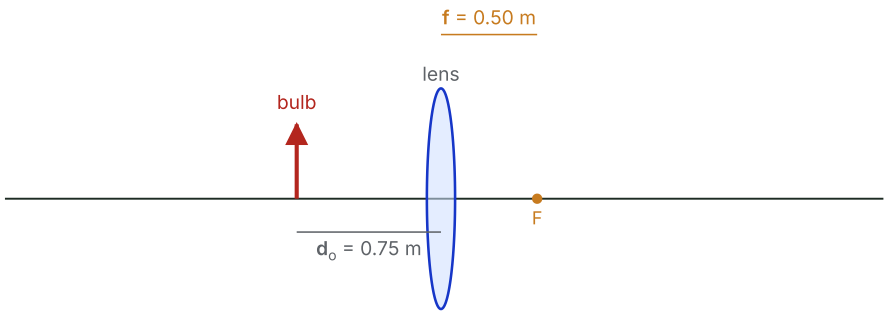


Figure 2. The bulb, the lens and the screen. The bulb and the focal length are given; **where the screen goes is what the equation decides.**

THE PUNCHLINE

$R = 2f$, and every equation from Note 6 still holds

Back to the prediction: a 40 cm radius gives a **20 cm** focal length, not 40. Mirrors bring no new physics into this module — only new signs, and one factor of two.

Looking ahead

Everything so far has used one optic at a time. **Note 9** puts two in a row, and the whole trick is in the footnote to the table above: *the image of the first becomes the object of the second*.

2. Which sign does f take for a **convex** mirror?

○ Positive, as for all mirrors. ○ Negative. ○ Positive if the object is real, negative otherwise. ○ Convex mirrors have no focal length.

3. A calculation for a mirror returns $d_i = -15$ cm. Where is the image?

○ 15 cm in front of the mirror; it is real. ○ 15 cm behind the mirror; it is virtual. ○ At the centre of curvature. ○ There is no image.

4. An object is 15 cm in front of a concave mirror of radius 20 cm. Find the magnification m.

Check Clear

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